

IoTfy Services Details:

Client: Power2GO



Product Developed: WallHub – Power2GO

- **Category:** Intelligent Solar Energy & Battery Management System
- **Target Audience:** Homes, offices, small businesses, and off-grid or semi-grid environments facing unreliable power.
- **Collaboration with IoTfy Solutions:**
 - **Hardware & System Architecture:** IoTfy Solutions engineered an all-in-one solar hub featuring a 580W solar panel integrated with MPPT charging, a dual 24V battery architecture, and a compact custom PCB (5×3 inches).
 - **Smart Load & Relay Control:** Implemented MOSFET-based load switching to support up to three 24V LED lights, a 12V fan, and USB/mobile charging ports, controllable via onboard buttons or IR remote.
 - **Battery Protection Logic:** Developed automated relay-based battery switching (activates when charge drops below 10%) and parallel charging to prevent deep discharge and maximize battery lifespan.
 - **Real-Time OLED Monitoring:** Integrated a 10-second cycling display showing runtime estimations (days/hours/minutes left), battery health, active load icons, and MPPT/AC charging status.
 - **IoT Readiness:** Powered by an ESP32-ready micro-controller platform enabling future Wi-Fi and Bluetooth cloud connectivity.

Client: K-Electric DNA



- **Company Description:** Pakistan's primary integrated power utility serving Karachi.

Product Developed: IOTFIY PoleKit

- **Category:** Smart Infrastructure Safety & IoT Electrical Monitoring System
- **Target Audience:** Electric utilities, municipal authorities, smart cities, housing societies, and industrial parks.
- **Collaboration with IoTfiy Solutions:**
 - **Hardware & Sensor System:** IoTfy Solutions developed an all-weather, rugged hardware kit mounted directly on utility poles to continuously monitor voltage (V), current (A), load behavior, and dangerous current leakage—preventing electrocution risks during rain or water ingress.
 - **MERN Stack Cloud Dashboard:** Built a full-stack centralized web dashboard (React.js, Express.js, Node.js, MongoDB) featuring live status tracking, historical performance logging, and real-time alerts.

- **Interactive Map & GIS Integration:** Implemented interactive map-based GPS pole tracking organized by area, street, and zone for rapid fault localization and emergency team dispatch.
- **Role-Based Access Control (RBAC):** Designed a three-tier access architecture (**Admin, Manager, and User**) to manage areas, streets, device registrations, and user permissions securely.
- **Predictive Maintenance:** Engineered data-logging capabilities to analyze aging or repeatedly faulty poles, enabling proactive maintenance scheduling rather than reactive incident response.

Client: Getz Pharma



- **Company Description:** Getz Pharma is a leading multinational pharmaceutical corporation and the largest pharmaceutical company in Pakistan, operating across more than 40 countries.

Product Developed: Enterprise Social Engagement Platform

- **Category:** Enterprise Mobile/Web Social Media & Event Management Platform
- **Target Audience:** Getz Pharma corporate employees, HR teams, event organizers, and executive leadership.
- **Collaboration with IoTfy Solutions:**
 - **Dedicated Enterprise Social Ecosystem:** IoTfy Solutions built a custom, secure private social network for Getz Pharma to centralize communication and boost engagement for large-scale corporate gatherings (such as their Annual Sales Event serving 5,000+ employees).
 - **Interactive Media & Gamification:** Designed dynamic feed features—including post/photo/video sharing, stories, polls, quizzes, and likes/comments—paired with a point and badge gamification system to incentivize active participation.
 - **Centralized Broadcast & Notifications:** Integrated admin-controlled official announcements, agenda updates, venue maps, and real-time push notifications to eliminate fragmented communication channels.
 - **Peer Mentorship & Knowledge Modules:** Built interactive sub-modules for document/video training uploads, course certifications, and peer-to-peer QA forums with helper point rewards.
 - **Feedback & Real-Time Analytics:** Incorporated an anonymous feedback submission portal along with a backend analytics dashboard to measure employee participation metrics, popular content, and active user insights.

Client: GameNest



- **Company Description:** GameNest is an automated entertainment and arcade operator modernizing gaming venues and commercial vending operations through smart IoT technology.

Product Developed: IoT-Based Arcade & Vending Machine Automation Platform

- **Category:** Industrial IoT Hardware, Edge Computing & Automated Payment Infrastructure
- **Target Audience:** Arcade operators, vending machine vendors, franchise owners, and multi-location entertainment venues.
- **Collaboration with IoTfy Solutions:**
 - **Edge Hardware & Gateway Development:** Installed ESP32 microcontrollers inside legacy machines for real-time coin pulse detection alongside Raspberry Pi gateways for local data aggregation and processing.
 - **Contactless Digital Payment Integration:** Integrated NFC and QR code scanners supporting Apple Pay, Google Pay, and Bambora API to eliminate cash dependency and prevent revenue leakage.
 - **AWS Cloud & Real-Time Dashboard:** Engineered a web portal (React/Next.js backed by AWS IoT Core, Lambda, and DynamoDB) providing live revenue logs, machine heatmaps, peak usage analytics, and multi-venue grouping.
 - **Over-The-Air (OTA) & Maintenance Alerts:** Built remote firmware update capabilities (OTA) and automated system alerts (e.g., hopper jams, offline devices), reducing onsite technician visits by 40%.
 - **Role-Based Management:** Designed custom RBAC access levels for Admins, Venue Managers, and Field Technicians to streamline multi-location oversight.

Client: SSUET (Sir Syed University of Engineering and Technology)



- **Company Description:** SSUET is a leading engineering and technology university in Pakistan, dedicated to academic excellence, digital innovation, and advanced research.

Product Developed: IOTFIY Virtual Tour

- **Category:** 3D Interactive Virtual Campus Tour
- **Target Audience:** Prospective students, university visitors, faculty, international scholars, and administrative stakeholders.
- **Collaboration with IoTfy Solutions:**
 - **3D Virtual Campus Walkthrough:** Engineered high-fidelity 3D virtual environments and 360° interactive walkthroughs of SSUET's campus facilities, departments, labs, and key infrastructure.
 - **Interactive Digital Asset Modeling:** Modeled custom 3D assets, floor plans, and campus locations to allow seamless digital navigation without physical location constraints.

- **Web Integration & Accessibility:** Integrated the interactive 3D virtual tour directly into the web interface, allowing visitors to explore the university grounds remotely 24/7 across desktop and mobile devices.

Client: Swedish Restaurant Group

Service Provider: IOTFIY SOLUTIONS

Project Type: Enterprise IoT Hardware + Cloud Monitoring Platform

Project Description (Enhanced)

FoodTrace is an enterprise-grade, end-to-end IoT cold chain monitoring solution designed to protect restaurant groups from catastrophic inventory loss, food safety violations, and operational blind spots.

High-volume restaurants rely heavily on uninterrupted refrigeration. A single unnoticed freezer failure can destroy inventory worth **\$10,000+ overnight**, damage brand reputation, and lead to serious regulatory penalties. FoodTrace eliminates this risk by delivering **real-time visibility, predictive alerts, and centralized governance** across all cold storage units.

Built by **IOTFIY SOLUTIONS**, FoodTrace transforms cold chain management from a reactive, manual process into a **proactive, data-driven, and automated system**—ensuring food safety, business continuity, and compliance at scale.

1. The Solution: Industrial IoT Ecosystem

A. Custom IoT Hardware Layer

Purpose-built hardware designed for commercial kitchen environments:

- **Dual-Zone Temperature Monitoring**
Simultaneous tracking of:
 - Ambient (Room) Temperature
 - Internal Freezer Temperature
- **Industrial-Grade Sensors**
 - Extreme cold resistance
 - High accuracy for food safety compliance
- **Device Health Intelligence**
 - Low battery detection
 - Sensor connectivity monitoring
 - Hardware failure alerts

B. Real-Time Alert & Protection System

FoodTrace eliminates silent failures with instant alerts:

- Custom temperature thresholds
- Instant anomaly detection
- Alerts via mobile & desktop
- Escalation for critical failures

This allows staff to **move inventory, fix equipment, or intervene immediately**—before losses occur.

2. Centralized Governance Dashboard

The cloud dashboard acts as a **command center** for the entire restaurant group.

Key Capabilities:

- **Live Multi-Location View**
Monitor all refrigerators and freezers across venues in real time
- **Historical Analytics & Logs**
 - Temperature trends
 - Failure patterns
 - Compliance reporting
- **Multi-Tier Management**
 - Organization-level oversight
 - Venue-level control
 - Device-level monitoring
- **OTA (Over-The-Air) Firmware Updates**
Remotely upgrade device software without site visits
- **Role-Based Access Control (RBAC)**
 - Regional managers oversee multiple sites
 - Local staff access only their kitchen units

3. Strategic Business Benefits

3.1 Financial Protection

- Prevents \$10,000+ inventory losses per incident
- Reduces emergency repair costs
- Lowers insurance and compliance risks

3.2 Food Safety & Compliance

- Continuous cold chain integrity
- Audit-ready temperature reports
- Reduced risk of license suspension

3.3 Operational Excellence

- No manual temperature logging
- Reduced staff workload

- Faster maintenance decisions

3.4 Scalable Growth

- Easily onboard new restaurant locations
- Centralized monitoring across regions
- Cloud-native scalability

4. Why Choose IOTFIY SOLUTIONS

- Proven expertise in **industrial IoT systems**
- Custom hardware + cloud integration
- Enterprise-grade security & scalability
- Long-term support & OTA lifecycle management
- Domain understanding of food & compliance industries

5. Conclusion

FoodTrace is not just an IoT project—it is a **business-critical safety net** for modern restaurant operations.

By partnering with **IOTFIY SOLUTIONS**, FoodTrace enables restaurant groups to:

- Protect revenue
- Maintain compliance
- Eliminate blind spots
- Scale with confidence

FoodTrace ensures that **no freezer failure ever goes unnoticed again.**

Client: Insight Securities



• **Company Description:**

Insight Securities is a leading financial services and brokerage firm providing investment, trading, and capital market solutions while operating highly secure and mission-critical IT infrastructure.

Product Developed: **Data Center Environmental Monitoring System (EMS)**

• **Category:** Enterprise Data Center Infrastructure Monitoring & Environmental Intelligence Platform

• **Target Audience:** IT managers, data center operators, infrastructure teams, and facility management personnel.

• **Collaboration with IoTfy Solutions:**

- **Enterprise Environmental Monitoring:** Designed and deployed an industrial-grade IoT monitoring system to continuously track temperature, humidity, smoke detection, water leakage, and other critical environmental parameters within the data center to ensure uninterrupted operations.
- **Real-Time Alert & Incident Management:** Engineered intelligent threshold-based alerts with instant notifications for abnormal environmental conditions, enabling rapid response before equipment failures or service disruptions occur.
- **Centralized Monitoring Dashboard:** Developed a cloud-based monitoring dashboard providing live environmental status, historical trends, device health monitoring, and comprehensive reporting for proactive infrastructure management.
- **Predictive Infrastructure Protection:** Implemented historical analytics and continuous monitoring capabilities to help identify environmental risks early, reducing downtime, protecting critical IT assets, and improving operational reliability.

Client: Pakistan State Oil (PSO)



Pakistan State Oil

• **Company Description:**

Pakistan State Oil (PSO) is Pakistan's largest energy company, operating nationwide fuel storage terminals, laboratories, and industrial facilities with strict safety and environmental compliance requirements.

Product Developed: **Smart Fume Hood Controller**

• **Category:** Industrial Laboratory Safety Automation & Environmental Control System

• **Target Audience:** Laboratory managers, HSE departments, research facilities, refinery laboratories, and industrial safety engineers.

• **Collaboration with IoTfy Solutions:**

- **Intelligent Fume Hood Controller Design:** Designed and developed a custom embedded controller to automate laboratory fume hood operations while maintaining safe airflow and environmental conditions.
- **Safety-Based Automation Logic:** Engineered intelligent control algorithms to regulate ventilation based on operating conditions, improving laboratory safety while minimizing unnecessary energy consumption.
- **Real-Time Equipment Monitoring:** Integrated continuous operational monitoring, equipment status indication, and fault detection to ensure reliable fume extraction and rapid maintenance response.
- **Energy-Efficient Laboratory Operations:** Optimized airflow management and operational scheduling to reduce power consumption while maintaining compliance with laboratory safety standards and environmental regulations.

Client: Invo Creator

- **Company Description:**

Invo Creator is a technology-driven organization focused on developing innovative digital platforms and smart business solutions for enterprise operations.

Product Developed: **IoT-Based Environmental Monitoring Application**

- **Category:** IoT Software Platform & Environmental Monitoring System

- **Target Audience:** Commercial facilities, industrial organizations, smart offices, and enterprise facility managers.

- **Collaboration with IoTfy Solutions:**

- **Cross-Platform IoT Application Development:** Designed and developed a modern mobile and web application enabling centralized monitoring of multiple IoT environmental monitoring devices from a single interface.
- **Real-Time Sensor Visualization:** Integrated live monitoring for temperature, humidity, air quality, and other environmental parameters with intuitive dashboards, charts, and device status indicators.
- **Intelligent Alerts & Notifications:** Implemented configurable threshold-based notifications to instantly inform users about abnormal environmental conditions requiring immediate attention.
- **Cloud-Based Device Management:** Built scalable cloud infrastructure supporting secure device onboarding, remote monitoring, historical analytics, and multi-user access management for enterprise deployments.

Client: Indus Hospital



- **Company Description:**

The Indus Hospital is one of Pakistan's leading healthcare institutions, providing advanced medical services through modern healthcare technologies and critical care infrastructure.

Product Developed: **Shared Ventilator Flow Analyzer**

- **Category:** Medical Device Testing & Respiratory System Measurement Solution
- **Target Audience:** Biomedical engineers, hospital technical teams, intensive care units (ICUs), and medical equipment manufacturers.
- **Collaboration with IoTfiy Solutions:**
 - **Precision Flow Measurement System:** Designed and developed a high-accuracy flow analyzer capable of measuring airflow characteristics for shared ventilator configurations under varying operating conditions.
 - **Medical Device Performance Validation:** Engineered testing capabilities to evaluate airflow distribution, pressure stability, and ventilation performance to support safe and reliable ventilator operation.
 - **Real-Time Data Acquisition:** Integrated continuous measurement and visualization of respiratory flow parameters, enabling engineers to monitor system performance during testing and calibration.
 - **Engineering Support for Critical Healthcare:** Delivered a reliable validation platform assisting biomedical teams in optimizing shared ventilator performance, improving testing efficiency, and supporting critical care research initiatives.

Client: LuckyOne Mall



- **Company Description:** LuckyOne Mall is one of South Asia's largest shopping malls and lifestyle destinations, serving thousands of daily visitors through hundreds of retail stores, restaurants, entertainment facilities, and public spaces.

Product Developed: **IOTFIY Smart Building Ecosystem Prototype**

- **Category:** Enterprise Smart Building IoT, Environmental Intelligence & Facility Management Platform
- **Target Audience:** Mall operations executives, facility managers, HVAC engineers, security teams, and building maintenance departments.
- **Collaboration with IoTfiy Solutions:**

- **Multi-Zone Environmental Monitoring:** Developed and deployed prototype IoT devices to continuously monitor indoor and outdoor air quality, public washroom odor levels, temperature, and humidity across multiple areas of the mall.
- **Smart Food Court Safety Monitoring:** Engineered intelligent gas leakage detection devices for food court operations, providing real-time alerts to improve visitor safety and reduce operational risks.
- **Unified IOTFIY Ecosystem Dashboard:** Built a centralized cloud dashboard aggregating live data from all deployed IoT devices—including odor monitoring, air quality sensors, gas leakage detectors, and environmental monitoring systems—into a single operational interface.
- **Smart Facility Intelligence:** Enabled facility management teams with real-time environmental visibility, historical analytics, proactive maintenance insights, and centralized monitoring to improve operational efficiency and enhance the overall visitor experience.